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Section: B(SE)-1D

GitHub Repo: https://github.com/m-huzaifahafeez/M.Huzaifa-Hafeez_26K-3117_PF_LAB-all-

Task1:

```
#include <stdio.h>
int main()
{
    int t;
    float fare,d;
    printf("\nenter distance in km");
    scanf("%f",&d);
    printf("\nenter what hour it is 0-23");
    scanf("%d",&t);
    if(d<=0)
    {
        printf("\ninvalid distance input");
    }
    else
    {
        if(t<6 || t>22)
        {
            fare=50.00+40.00+((d-1)*22.00);
        }
        else
        {
            fare=50.00+((d-1)*22.00);
        }
        printf("\n your fare is %.2f Rs.",fare);
    }
}
```

Problems Output Debug Console Terminal Ports

enter distance in km22

enter what hour it is 0-234

your fare is 552.00 Rs.

PS C:\Users\Huzaifa\mycode(PF_lab_all> █

Task2:

```
#include<stdio.h>
int main()
{

    float bill, disc_rate = 0.0, disc_amt, final_amt;
    int is_member;

    printf("Enter total bill amount: ");
    scanf("%f", &bill);

    printf("Enter membership status (1 for member, 0 for non-member): ");
    scanf("%d", &is_member);

    if (bill < 500) {
        disc_rate = 0.0;
    } else if (bill >= 500 && bill <= 1999) {
        if (is_member == 1) {
            disc_rate = 0.10;
        } else {
            disc_rate = 0.05;
        }
    } else if (bill >= 2000) {
        if (is_member == 1) {
            disc_rate = 0.15;
        } else {
            disc_rate = 0.08;
        }
    }
}

disc_amt = bill * disc_rate;
final_amt = bill - disc_amt;

printf("Discount Amount: Rs %.2f\n", disc_amt);
printf("Final Payable Amount: Rs %.2f\n", final_amt);

return 0;
}
```

Problems Output Debug Console Terminal Ports

```
Enter total bill amount: 2500
Enter membership status (1 for member, 0 for non-member): 1
Discount Amount: Rs 375.00
Final Payable Amount: Rs 2125.00
PS C:\Users\Huzaifa\mycode(PF_lab_all> [
```

Task3:

```
#include <stdio.h>

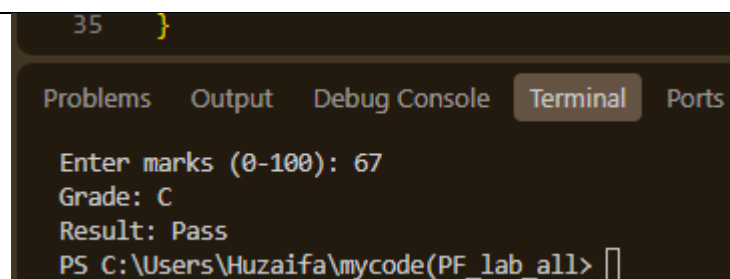
int main() {
    int m;

    printf("Enter marks (0-100): ");
    scanf("%d", &m);

    if (m < 0 || m > 100) {
        printf("Invalid Marks\n");
        return 0;
    }

    if (m >= 90) {
        printf("Grade: A+\n");
        printf("Result: Pass\n");
    } else if (m >= 80) {
        printf("Grade: A\n");
        printf("Result: Pass\n");
    } else if (m >= 70) {
        printf("Grade: B\n");
        printf("Result: Pass\n");
    } else if (m >= 60) {
        printf("Grade: C\n");
        printf("Result: Pass\n");
    } else if (m >= 50) {
        printf("Grade: D\n");
        printf("Result: Pass\n");
    } else {
        printf("Grade: F\n");
        printf("Result: Fail\n");
    }

    return 0;
}
```



35 }

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Enter marks (0-100): 67
Grade: C
Result: Pass
PS C:\Users\Huzaiifa\mycode(PF_lab_all> [

Task4:

```
#include <stdio.h>

int main() {
    int cap, cur;
    float rate, req_liters, time;
    int mins;
    float cost;

    printf("Enter tank capacity (liters): ");
    scanf("%d", &cap);

    printf("Enter current water level (liters): ");
    scanf("%d", &cur);

    printf("Enter fill rate (liters per min): ");
    scanf("%f", &rate);

    if (cur >= cap) {
        printf("Tank Already Full\n");
        return 0;
    }

    req_liters = cap - cur;
    time = req_liters / rate;

    mins = (int)time;
    if (time > mins) {
        mins = mins + 1;
    }

    cost = mins * 3.50f;

    printf("Required Time: %.2f minutes\n", time);
    printf("Billed Electricity Cost: Rs %.2f\n", cost);

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
Enter tank capacity (liters): 150
Enter current water level (liters): 70
Enter fill rate (liters per min): 3
Required Time: 26.67 minutes
Billed Electricity Cost: Rs 94.50
PS C:\Users\Huzaifa\mycode(PF_lab_all> 
```

Task5:

```
#include <stdio.h>

int main() {
    float amt, rate = 0.0, bonus, total;
    int net, wknd;

    printf("Enter load amount: ");
    scanf("%f", &amt);

    printf("Enter network code (1=Jazz, 2=Telenor, 3=Ufone): ");
    scanf("%d", &net);

    printf("Enter weekend status (1=weekend, 0=weekday): ");
    scanf("%d", &wknd);

    if (amt < 100) {
        rate = 0.0;
    } else if (amt >= 100 && amt <= 499) {
        if (wknd == 1) {
            if (net == 3) {
                rate = 0.05;
            } else {
                rate = 0.10;
            }
        } else {
            rate = 0.05;
        }
    } else if (amt >= 500) {
        if (net == 1 || wknd == 1) {
            rate = 0.20;
        } else {
            rate = 0.12;
        }
    }

    bonus = amt * rate;
    total = amt + bonus;

    printf("Bonus Amount: Rs %.2f\n", bonus);
    printf("Final Loaded Balance: Rs %.2f\n", total);

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
Enter load amount: 350
Enter network code (1=Jazz, 2=Telenor, 3=Ufone): 1
Enter weekend status (1=weekend, 0=weekday): 0
Bonus Amount: Rs 17.50
Final Loaded Balance: Rs 367.50
PS C:\Users\Huzaifa\mycode(PF_lab_all> █
```

Task6:

```
#include <stdio.h>

int main() {
    int days, book_type, is_priority;
    double fine = 0.0;
    printf("Enter number of overdue days: ");
    scanf("%d", &days);
    printf("Enter book type (1 = Regular, 2 = Reference, 3 = Rare): ");
    scanf("%d", &book_type);
    printf("Priority member? (1 = Yes, 0 = No): ");
    scanf("%d", &is_priority);
    switch (book_type) {
        case 1:
            if (days <= 7) {
                fine = days * 5.0;
            } else {
                fine = (7 * 5.0) + ((days - 7) * 10.0);
            }
            break;

        case 2:
            fine = days * 15.0;
            break;

        case 3:
            fine = days * 30.0;
            break;

        default:
            printf("Invalid book type.\n");
            return 1;
    }
    if (is_priority == 1 && book_type != 3) {
        fine *= 0.80; // Apply 20% discount
    }
    printf("\nTotal Fine: Rs %.2f\n", fine);

    if (book_type == 3 && days > 10) {
        printf("Banned from Borrowing\n");
    }
}
```

```
return 0;
}
```

Problems Output Debug Console **Terminal** Ports

```
Enter number of overdue days: 4
Enter book type (1 = Regular, 2 = Reference, 3 = Rare): 3
Priority member? (1 = Yes, 0 = No): 1

Total Fine: Rs 120.00
PS C:\Users\Huzaiifa\mycode(PF lab all> █
```

Task7:

```
#include <stdio.h>
```

```
int main() {
    float avg;
    int m, fit;
```

```
    printf("Enter batting average: ");
    scanf("%f", &avg);
```

```
    printf("Enter matches played: ");
    scanf("%d", &m);
```

```
    printf("Enter fitness failure status (1=failed, 0=passed): ");
    scanf("%d", &fit);
```

```
    if (m < 5) {
        printf("Rejected - Insufficient Matches\n");
    } else if (avg >= 35.0f && m >= 10) {
        printf("Selected\n");
    } else if (avg >= 25.0f && avg <= 34.99f && m >= 20) {
        if (fit == 1) {
            printf("Rejected - Fitness\n");
        } else {
            printf("Selected (Experience Quota)\n");
        }
    } else {
        printf("Not Selected\n");
    }
}
```

```
return 0;
}
```

Problems Output Debug Console Terminal Ports

```
Enter batting average: 67
Enter matches played: 420
Enter fitness failure status (1=failed, 0=passed): 0
Selected
PS C:\Users\Huzaifa\mycode(PF_lab_all> █
```

Task8:

```
#include <stdio.h>

int main() {
    int cat, type;
    float bill, rate_s = 0.0, rate_d = 0.0;
    float charge, disc, total;

    printf("Enter Meal Category (1 = Fast Food, 2 = Desi Food, 3 = Chinese): ");
    scanf("%d", &cat);

    printf("Enter Bill Amount: ");
    scanf("%f", &bill);

    printf("Enter Customer Type (1 = Student, 2 = Regular): ");
    scanf("%d", &type);

    switch (cat) {
        case 1:
            rate_s = 0.05;
            break;
        case 2:
            rate_s = 0.08;
            break;
        case 3:
            rate_s = 0.10;
            break;
        default:
            printf("Invalid Selection\n");
            return 0;
    }

    if (type == 1) {
        if (bill >= 1000) {
            rate_d = 0.15;
        } else {
            rate_d = 0.05;
        }
    } else if (type == 2) {
        if (bill >= 1000) {
            rate_d = 0.10;
```

```

    } else {
        rate_d = 0.00;
    }
} else {
    printf("Invalid Selection\n");
    return 0;
}

charge = bill * rate_s;
disc = bill * rate_d;
total = bill + charge - disc;

printf("Service Charge: Rs %.2f\n", charge);
printf("Discount: Rs %.2f\n", disc);
printf("Final Payable Amount: Rs %.2f\n", total);

return 0;
}

```

Problems Output Debug Console **Terminal** Ports

```

Enter Meal Category (1 = Fast Food, 2 = Desi Food, 3 = Chinese): 2
Enter Bill Amount: 3500
Enter Customer Type (1 = Student, 2 = Regular): 1
Service Charge: Rs 280.00
Discount: Rs 525.00
Final Payable Amount: Rs 3255.00
PS C:\Users\Huzaifa\mycode(PF_lab_all> 

```